Think Less, Act Better: Adaptive Reasoning for Vision-Language-Action Models

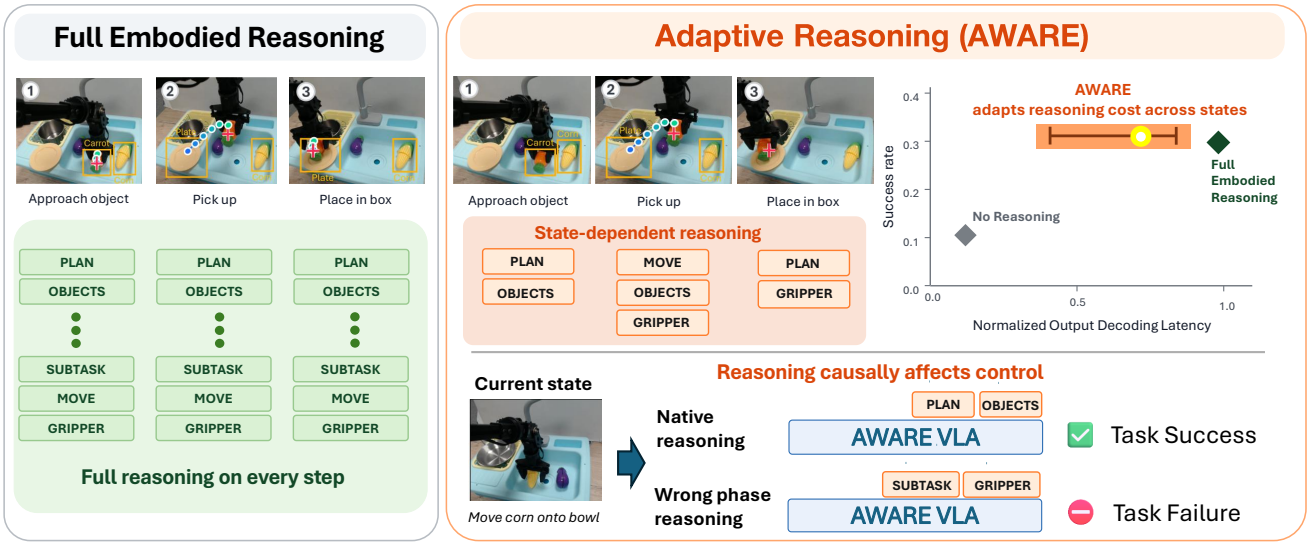


Fig. 1: **Adaptive embodied reasoning.** Unlike full embodied chain-of-thought reasoning, which generates the complete reasoning trace at every control step, AWARE adapts what and how much it reasons about to the current state. This preserves full-reasoning task performance at lower inference cost, while causal interventions show that changing the reasoning can substantially affect control success. The normalized output decoding latency denotes the time necessary to generate the full model output at inference, including reasoning and action tokens, normalized with respect to the Full Reasoning baseline.

***Abstract*— Robot chain-of-thought (CoT) reasoning can improve the performance and generalization of Vision-Language-Action (VLA) policies by generating intermediate reasoning before predicting actions. However, embodied CoT policies typically produce long, structured reasoning traces at every control step, substantially increasing inference latency even when many reasoning primitives are not relevant to the current state. To this end, we propose a novel offline approach for adaptive embodied reasoning that teaches a VLA when to reason, what to reason about and how much reasoning to generate at each control step. Starting from a full-reasoning policy, we use its own action predictions to identify compact, action-predictive reasoning traces and learn them through self-distillation while preserving action behavior. Extensive experiments across simulated benchmarks and real-world manipulation show that AWARE reduces generated reasoning primitives by up to 30% in simulation and by 40.9% on a real-world WidowX robot, while maintaining task success comparable to Full Reasoning. We further find that its reasoning changes systematically across manipulation phases, adapting both how much it reasons and what it reasons about to the current control state. Finally, causal interventions demonstrate that this reasoning directly affects control: using otherwise correct reasoning at the wrong manipulation phase drops task success from 88% to 46%.**

I. INTRODUCTION

Humans have the ability to adapt their reasoning to the situation at hand, acting quickly in familiar settings while reasoning more carefully in unfamiliar or difficult ones. This distinction between automatic and deliberate modes of thinking has a long history in cognitive science [1]–[4], and is commonly described today in terms of System 1 and System 2 reasoning [5], [6]. Importantly, this is not a binary choice: we can vary both how much we reason and what we reason about depending on the situation.

In contrast, current Vision-Language-Action (VLA) policies do not yet capture this flexibility. Standard VLAs predict actions directly from observations [7]–[9], while embodied chain-of-thought (CoT) policies [10]–[13] generate grounded intermediate reasoning before acting. Although such reasoning can improve performance and generalization, these policies typically generate a long, structured reasoning trace at every control step. The length of these traces can significantly increase inference latency, rendering the embodied CoT-based policies impractical for closed-loop control. Furthermore, the reasoning that is useful for one action might not be useful for the next: high-level planning may matter when deciding what to do, spatial reasoning while approaching an object, and gripper reasoning during contact. This raises a natural question: *can a VLA learn when to reason, what to reason about, and how much reasoning is needed for each action?*

Learning adaptive embodied reasoning requires more than deciding whether to reason: a policy must determine what to reason about and when that reasoning is needed. Existing approaches reduce reasoning cost by omitting explicit

reasoning [14], reusing it across control steps [15], or adapting latent computation [16]–[18]. These methods adapt the amount or frequency of computation, but do not explicitly treat the semantic composition of reasoning as a control-dependent decision. We instead study how a VLA can learn state-dependent reasoning *within trajectories*, entirely from offline robot demonstrations. Our key observation is that the relevance of a reasoning primitive can be estimated from its effect on action prediction. By selectively removing reasoning primitives and measuring the resulting action-prediction error, we obtain an action-grounded signal for what reasoning is needed at each control state.

Building on this observation, we introduce AWARE (**A**ction-a**W**are **A**daptive **R**easoning for **E**mbodied control), an offline approach for adaptive embodied reasoning. Starting from a VLA trained with full embodied reasoning, we evaluate different subsets of reasoning primitives and identify compact traces that best predict the demonstrated action. We then distill these state-dependent reasoning traces back into the policy while preserving the control behavior of the full-reasoning model. At inference time, the resulting VLA decides when to reason, what to reason about, and how much reasoning to generate directly from the current observation, without online reinforcement learning or an external verifier.

Our extensive evaluation across simulated and real-world manipulation shows that AWARE reduces generated reasoning primitives by up to 30% in simulation and by 40.9% on a real-world WidowX robot, while preserving Full Reasoning performance. We find that the learned reasoning changes systematically across control states and manipulation phases. More importantly, causal interventions show that these reasoning decisions directly affect control: using otherwise correct reasoning from the wrong manipulation phase reduces task success from 88% to 46.2%, while perturbing reasoning that is irrelevant to the current action has little effect. Together, these results show that effective embodied reasoning depends not only on how much a policy reasons, but also on *what* it reasons about and *when*.

II. RELATED WORK

A. Embodied Reasoning

Vision-Language-Action (VLA) models adapt pretrained vision-language models to robot control by predicting actions from visual observations and language instructions [7]–[9]. Leveraging large-scale vision-language pretraining and diverse robot data, VLAs have demonstrated strong generalization across tasks, scenes, and robot embodiments [12], [19], [20].

Inspired by chain-of-thought (CoT) reasoning in language models [21], [22], recent robot policies generate intermediate reasoning before predicting actions. ECoT [10], for example, autoregressively generates grounded reasoning primitives for task planning, spatial grounding, motion, and robot state. Subsequent work has explored hierarchical language motions [23], spatial reasoning [24], action-centric representations [25], and visual or spatial reasoning [18], [26]–[28]. While these works primarily study what intermediate reasonings should represent, they typically use a fixed reasoning structure at inference time. We instead study which reasoning is needed at each control state and how it should change throughout a trajectory.

B. Efficient and Adaptive Embodied Reasoning

Generating an autoregressive reasoning trace before every action can substantially increase inference cost. ECoT-Lite [14] develops training strategies that can omit explicit reasoning at inference, while Fast ECoT [15] exploits temporal redundancy by reusing reasoning across control steps. Other approaches adapt latent computation or selectively recompute intermediate representations [16]–[18]. These approaches reduce reasoning cost by changing whether, how often, or how deeply a policy reasons, but do not directly study how the semantic content of explicit reasoning should vary across control states.

Action prediction provides a complementary signal for what information matters to control. EyeRobot [29] uses it to learn where a robot should look, while R&B-EnCoRe [30] uses action-predictiveness to refine reasoning supervision. We use this signal to make reasoning relevance a state-dependent policy decision: a single VLA varies the amount and content of its explicit reasoning from one control state to the next. We further causally test whether it performs the right reasoning at the right time.

C. Reasoning Distillation

Reasoning distillation has been widely studied in language models, where teacher-generated reasoning traces provide supervision for a student [31]–[33]. Beyond distilling complete rationales, prior work has also explored retaining only their most informative parts [34]. Related ideas have been applied to robot policies by distilling explicit reasoning into compact latent representations [35] or transferring capabilities across VLA models [36], [37]. In contrast, we use self-distillation to teach the same VLA state-dependent reasoning while preserving full-reasoning action behavior.

III. METHOD

The goal of AWARE is to learn a VLA that reasons selectively, generating only the reasoning needed for the current action. Our approach consists of two stages. First, we identify state-dependent reasoning targets by selectively removing reasoning primitives from offline demonstrations and measuring their effect on action prediction. Second, we distill these targets back into the full-reasoning policy while preserving its action behavior. At inference time, AWARE decides what and how much to reason about directly from the current observation.

We consider autoregressive reasoning VLAs that generate a structured sequence of reasoning primitives before predicting the action in the same sequence. Given an offline robot demonstration dataset $\mathcal{D} = \{(x_t, a_t^*, y_t^{\text{full}})\}$, x_t denotes the current observation and language instruction, a_t^* the demonstrated action, and $y_t^{\text{full}} = (r_t^1, \ldots, r_t^K)$ the full reasoning

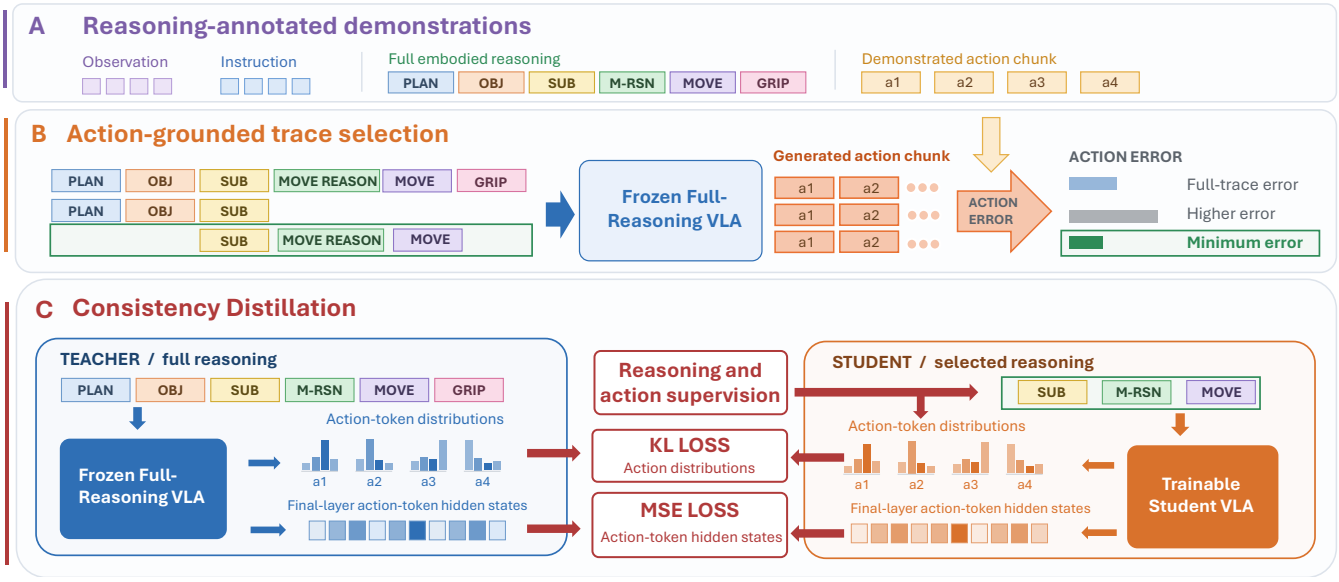


Fig. 2: Selecting shorter reasoning traces and consistency distillation. (A) Our method is initiated with a reasoning dataset used to post-train an autoregressive VLA to generate intermediate reasoning before action prediction. (B) A frozen full-reasoning policy scores candidate masks by action error. The action-first rule minimizes error and uses trace length only to break exact ties. (C) The student learns the selected trace and action while matching the full-trace teacher at action-token logits and hidden states.

trace composed of K semantic reasoning primitives. A subset of these primitives defines a candidate reasoning trace y_t^m, where $m \in \{0,1\}^K$ indicates which primitives are retained.

A. Action-Grounded Reasoning Selection

Our key idea is to use action prediction as a task-grounded signal for determining which reasoning is relevant in each state. For every demonstration sample $(x_t, a_t^*, y_t^{\text{full}})$, we construct a set of candidate masks $\mathcal{M}_t$ that retain different subsets of the K reasoning primitives. Each mask $m \in \mathcal{M}_t$ defines a candidate trace y_t^m, and the unmodified full-reasoning trace is always included as a candidate.

For the benchmark with $K = 7$ primitives, we evaluate a random sample of 30 masks in $\mathcal{M}_t$ along with empty and full trace using the frozen full-reasoning policy. For the benchmark with $K = 3$, we evaluate all candidate subsets. Conditioned on the candidate reasoning trace, the policy predicts an action chunk, which we compare to the demonstrated action:

$$e_t(m) = \|\text{Decode}(\pi_0(a \mid x_t, y_t^m)) - a_t^*\|_1 . \tag{1}$$

Intuitively, if removing a reasoning primitive increases the action-prediction error, that primitive contains information relevant to the current action. Conversely, reasoning that can be removed without degrading action prediction is a natural candidate to omit.

We select the candidate with the lowest action-prediction error. Due to the discrete action token prediction setup of this paper, ties in error occur frequently. We use trace length solely to break these exact ties:

$$m_t^* = \arg\min_{m \in \mathcal{M}_t} \left(e_t(m), n_{\text{tok}}(y_t^m)\right), \tag{2}$$

where the tuple is ordered lexicographically. Thus, action prediction takes priority over reasoning reduction: a shorter trace is preferred only when it achieves the same minimum action error. The selected trace $y_t^* = y_t^{m_t^*}$ provides state-specific supervision for what the policy should reason about at control step t.

Applying this procedure to the offline demonstration dataset produces $\mathcal{D}_{\text{sel}} = \left\{\left(x_t, a_t^*, y_t^{\text{full}}, y_t^*\right)\right\}$, which pairs each full reasoning trace with an action-grounded, state-dependent reasoning target. Because this selection is performed independently at every control state, the same reasoning primitive may be retained for one action and omitted for the next.

B. Consistency Distillation

The selected traces provide supervision for *what* the policy should reason about in each state, but learning these traces with supervised fine-tuning alone does not guarantee that the resulting shorter reasoning preserves the action behavior of the full-reasoning policy. We use consistency distillation to teach the policy state-dependent reasoning while anchoring its actions to those produced with the full trace.

We initialize both a frozen teacher T and a trainable student S_θ from the same full-reasoning policy π_0. For each tuple $(x_t, a_t^*, y_t^{\text{full}}, y_t^*) \in \mathcal{D}_{\text{sel}}$, the teacher is conditioned on the full reasoning trace y_t^{full}, while the student is trained on the selected trace y_t^*. The student is optimized with

$$\begin{aligned}\mathcal{L}_{\text{CD}} = \mathcal{L}_{\text{reason}}(y_t^*) + \lambda_{\text{act}}\mathcal{L}_{\text{act}}(a_t^*) \\ + \lambda_{\text{KL}}\mathcal{L}_{\text{KL}}\left(T(\cdot \mid x_t, y_t^{\text{full}}) \,\|\, S_\theta(\cdot \mid x_t, y_t^*)\right) \\ + \lambda_{\text{hid}}\mathcal{L}_{\text{hid}}\left(T(x_t, y_t^{\text{full}}), S_\theta(x_t, y_t^*)\right).\end{aligned} \tag{3}$$

Here, $\mathcal{L}_{\text{reason}}$ supervises the selected reasoning trace and $\mathcal{L}_{\text{act}}$ retains supervision from the demonstrated action. The KL term matches the student's action-token distribution to that of the full-reasoning teacher, while $\mathcal{L}_{\text{hid}}$ aligns their action-token hidden representations. Together, these objectives encourage the student to change *how it reasons* while preserving *how it acts*.

This procedure can be viewed as self-distillation: both teacher and student start from the same reasoning-capable VLA, but the teacher acts after the complete reasoning trace whereas the student learns to reach the same action behavior using state-dependent reasoning. After training, the teacher is discarded. At inference time, AWARE generates its selected reasoning and action autoregressively in a single policy call, without candidate evaluation or an external routing model.

C. Implementation Details

To preserve the autoregressive reasoning schema, omitted reasoning primitives retain their field boundaries while their content is replaced by a single atomic $\langle \text{skip} \rangle$ token. This preserves the structured layout of tags and the overall layout while allowing the policy to explicitly learn whether to generate each primitive. For consistency distillation, the KL objective is applied at the action-token positions and the representation loss matches final-layer action-token hidden states. We use $\lambda_{\text{act}} = 1$, $\lambda_{\text{KL}} = 1$, and $\lambda_{\text{hid}} = 1$ in all experiments. We use the autoregressive VLA architectures and training hyperparameters of the corresponding Full Reasoning baselines [10], [38], [39] unless stated otherwise.

IV. Experimental Results

Our experiments provide an empirical analysis of AWARE, evaluating adaptive embodied reasoning along several axes:

1) Can AWARE reduce reasoning and reasoning-decoding latency while preserving the task performance of a full-reasoning policy?
2) Does the learned policy adapt what it reasons about across control states and how much?
3) Does the selected reasoning causally affect control, and does performing the right reasoning at the right time matter?
4) Is the proposed approach applicable to real-world robot manipulation?

a) Evaluation setups: We evaluate AWARE across two simulated manipulation benchmarks and a real-world robot setting, spanning different robot embodiments, reasoning schemas, and VLA policies.

LIBERO-90. We evaluate MiniVLA [39] on LIBERO-90 [40], which simulates 90 language-conditioned manipulation tasks with a Franka Panda robot. The reasoning annotations follow the ECoT-Lite [14] format with seven reasoning primitives. In addition to the standard task distribution, we evaluate robustness using LIBERO-PRO [41], which introduces object, environment, and object-position perturbations while preserving the underlying task semantics.

RoboCasa. We additionally evaluate MiniVLA on two RoboCasa [42] pick-and-place tasks using a Kinova Jaco robot. The reasoning traces contain three primitives describing object grounding, next motion, and gripper path. For out-of-distribution evaluation, we expand the target-placement region while keeping the task, robot, and remaining environment conditions fixed.

Real-World Manipulation. Finally, we evaluate adaptive reasoning during physical robot control using OpenVLA [8] on a WidowX robot. We use the seven-field reasoning format introduced by ECoT [10] and evaluate on four pick-and-place tasks. This setting tests whether state-dependent reasoning learned through our offline procedure also emerges during closed-loop control in the physical world.

Reasoning trace formats. For LIBERO-90 and WidowX, we use the seven-field reasoning schema from ECoT [10]: PLAN, VISIBLE OBJECTS, SUBTASK REASONING, SUBTASK, MOVE REASONING, MOVE, and GRIPPER POSITION. Following the Bridge V2 ECoT [10] annotations released for WidowX, GRIPPER POSITION is replaced by GRIPPER PATH, which denotes future gripper positions. The final ACTION field is emitted after the reasoning trace and is not counted as a reasoning primitive. For RoboCasa, we use the BAR-X [38] representation with three primitives: object bounding boxes, future gripper path, and next motion. Following ECoT [10], primitives are generated autoregressively in a single sequence before the action.

b) Comparisons: We compare AWARE against two natural baselines of embodied reasoning.

No Reasoning. The policy predicts actions directly without generating an intermediate reasoning trace. This baseline measures whether explicit reasoning is beneficial in the corresponding task setting.

Full Reasoning. The policy generates the complete structured reasoning trace before every action. We use the converged full-reasoning policy to initialize AWARE and as the reference for evaluating whether adaptive reasoning preserves its control performance.

c) Metrics and training protocol: We primarily report task success, together with the fraction of reasoning primitives omitted, the number of generated reasoning tokens, and per-query reasoning-decoding latency. To study adaptivity, we additionally measure primitive retention across different manipulation phases. Full-reasoning and no-reasoning VLA policies are trained until task performance plateaus, after which AWARE is post-trained from the converged full-reasoning checkpoint.

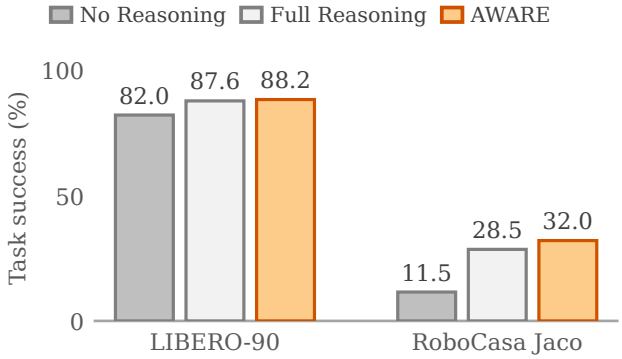


Fig. 3: **Adaptive reasoning preserves the benefits of Full Reasoning.** On in-distribution evaluations, AWARE remains close to Full Reasoning on both LIBERO-90 and RoboCasa, while retaining the large performance gain from explicit reasoning on RoboCasa (32.0% vs. 11.5% without reasoning).

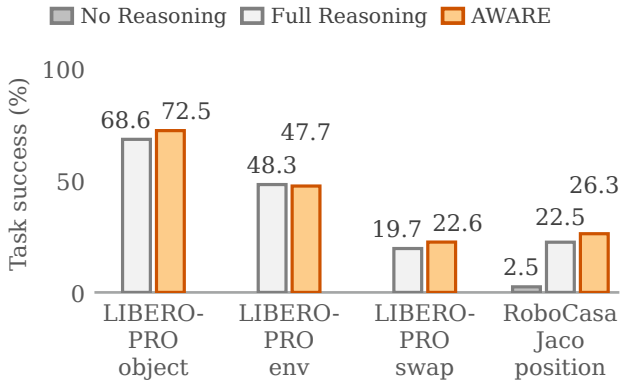


Fig. 4: **Adaptive reasoning remains competitive under distribution shift.** On LIBERO-PRO, AWARE matches Full Reasoning across object, environment, and position perturbations, and also improves performance on the RoboCasa target-position shift.

A. Adaptive Reasoning Performance and Efficiency

We first evaluate whether AWARE can reduce reasoning while preserving the control performance of the full-reasoning policy. As shown in Fig. 3, AWARE matches the Full-Reasoning on LIBERO-90 (88.2% vs. 87.6%) and achieves comparable success to Full-Reasoning on RoboCasa (32.0% vs. 28.5%). The RoboCasa results also highlight the benefit of explicit reasoning: performance drops to 11.5% when reasoning is removed entirely. Thus, AWARE retains the benefits of embodied reasoning while generating only a subset of the full reasoning trace.

This behavior persists under distribution shift (Fig. 4). On the LIBERO-PRO object, environment, and position-swap perturbations, AWARE achieves 72.5%, 47.7%, and 22.6% success, compared with 68.6%, 48.3%, and 19.7% for Full Reasoning. On the RoboCasa target-position shift, AWARE reaches 26.3% success, compared with 22.5% for Full Reasoning and only 2.5% without reasoning. These results suggest that selectively reducing reasoning does not compromise the generalization of the full-reasoning policy.

Crucially, this performance is achieved with substantially less reasoning. On LIBERO-90, AWARE skips 11.7% of the reasoning primitives; on RoboCasa, it skips 29.8%. This reduces reasoning content from 229.0 to 192.0 tokens per policy query (16.2%) on LIBERO-90 and from 108.0 to 76.3 tokens per query (29%) on RoboCasa. The shorter generations translate directly into lower reasoning-decoding latency, saving 87 ms per query on LIBERO-90 and 73 ms on RoboCasa while maintaining Full Reasoning baseline task success. Since the reasoning decision is made by the VLA itself, these savings require no external router or additional inference-time computation.

B. State-Dependent Reasoning

We next ask whether AWARE merely learns to generate less reasoning on average, or whether its reasoning changes with the current control state. To study this, we group policy queries by manipulation phase and measure how often each reasoning primitive is retained.

For the simulated LIBERO and RoboCasa analyses, we assign each query a phase automatically using simulator-detected grasp contact, the object's height relative to its initial position, its horizontal distance to the target, and its grasp or lift history. These labels are independent of the generated reasoning. In LIBERO, approach and grasp denote the absence and presence of a detected grasp before the first grasped lift of more than 2.5 cm above the object's initial height. After that lift, a query with the grasp maintained is labeled alignment, shown as Align in the table, when the object is within 8 cm of the goal in the horizontal plane and transport otherwise. Loss of grasp within that goal region is labeled release.

Phase contrast	Δ_{skip} (pp)	95% CI	Holm p
LIBERO – Mean over 7 primitives			
Align − Approach	+33.7	[23.9, 43.3]	0.0006
Align − Grasp	+47.5	[36.9, 58.5]	0.0006
Align − Transport	+32.4	[22.1, 42.9]	0.0006
LIBERO – MOVE			
Release − Transport	+67.9	[35.7, 96.4]	0.0265
RoboCasa – Next motion			
Release − Transport	+49.1	[27.5, 68.1]	0.0322
RoboCasa – Object boxes			
Transport − Grasp	−28.8	[−44.9, −11.8]	0.0382

TABLE I: **Phase-dependent skipping of reasoning primitives.** Selected significant phase contrasts are shown. Positive values mean more skipping in the left-hand phase. The LIBERO aggregate averages over seven reasoning primitives. Intervals are pointwise 95% confidence intervals; p-values are Holm-corrected across all 120 tested contrasts.

A reasoning primitive is one semantic field of the generated trace, distinct from a manipulation phase. LIBERO uses seven primitives whereas RoboCasa uses three primitives as described in IV. A primitive is skipped when its content is replaced by the skip token. For each table row, we subtract the skipping rate of the phase after the minus sign from that of the phase before it. For example, Align − Approach measures the alignment skipping rate minus the approach skipping rate.

Table I shows significant phase-dependent changes in learned reasoning behavior. On LIBERO-90, the overall skip rate increases by 47.5 percentage points from grasp to alignment, with similarly large increases from approach and transport to alignment. At the primitive level, MOVE reasoning is skipped 67.9 percentage points more often during release than during transport. We observe a similar pattern on RoboCasa, where next-motion reasoning is skipped 49.1 percentage points more often during release than during transport, while object-box reasoning is retained more often during transport than during grasp. We further investigate the average retention rates per phase. The RoboCasa curves show that next-motion reasoning is retained most frequently during transport and less frequently during release. Gripper reasoning follows a different pattern, with higher retention during grasp, alignment, and release than transport. Object-box retention also decreases from transport to alignment.

These results show that AWARE does more than learn a fixed compression of the full reasoning trace. Both the amount and content of its reasoning vary over the course

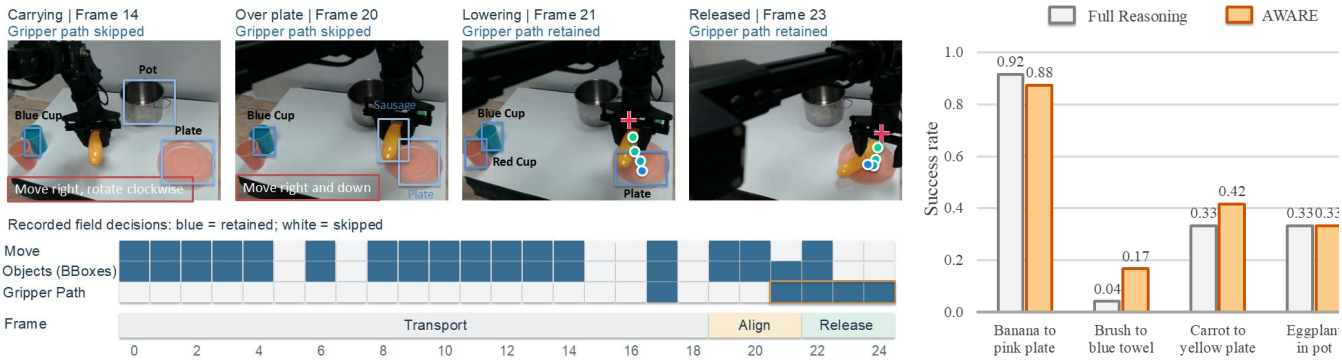


Fig. 5: **AWARE transfers adaptive reasoning to real-world manipulation. Left:** In a representative OpenVLA-ECoT rollout on WidowX, reasoning changes with the manipulation phase: gripper-path reasoning is omitted during transport and retained near alignment and release. The heatmap shows three representative primitives over 25 policy queries. **Right:** Across four real-world tasks, AWARE achieves task success comparable to Full Reasoning while reasoning selectively.

of control: different primitives become relevant at different stages of manipulation, and the same primitive can be retained for one action and omitted for another.

C. Causal Effects of Reasoning

We test whether the reasoning retained by AWARE causally affects control, rather than merely correlating with the current manipulation state. We compare the original policy with five interventions. First, we feed its unchanged reasoning back into the model and re-decode the action, testing whether the intervention procedure itself changes behavior. We then perturb either a frequently or rarely selected reasoning primitive while leaving the remaining reasoning unchanged. We reverse directional terms such as left and right, or mirror spatial coordinates. Finally, we replace the complete reasoning with valid reasoning generated by the same model in another episode of the same task, using reasoning from either the same or a different manipulation

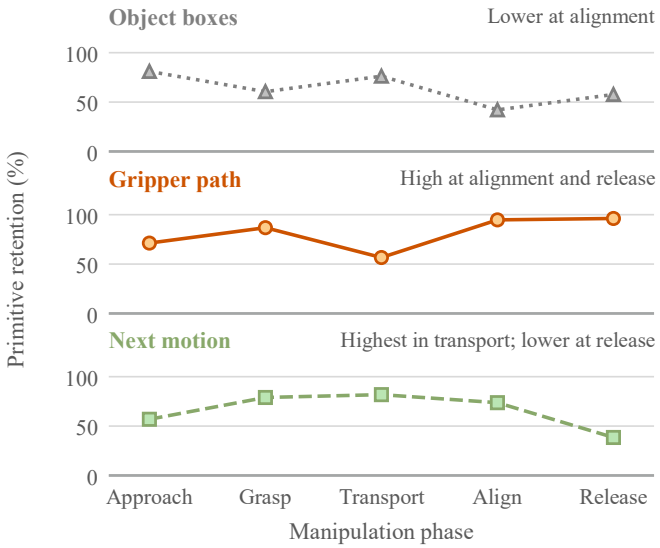


Fig. 6: **AWARE adapts what it reasons about across manipulation phases.** Average primitive-retention rates over 200 RoboCasa evaluation episodes across two tasks show distinct state-dependent reasoning patterns. In particular, next-motion reasoning is emphasized during transport and reduced at release, while gripper-path reasoning is retained most often during alignment and release.

phase. To isolate the effect of reasoning content from reasoning length, all intervened traces are padded or truncated to match the reasoning budget of the original policy query.

As shown in Fig. 7, re-decoding actions from the unchanged reasoning trace matches native execution, achieving 88.0% success on LIBERO-90 and 32.0% on RoboCasa. Corrupting a low-relevance primitive has little effect on LIBERO-90 (86.0% success), whereas corrupting a high-relevance primitive reduces success to 82.3%. We observe the same trend on RoboCasa, where success decreases from 11.0% for low-relevance corruption to 3.0% when action-relevant reasoning is corrupted. These results suggest that the reasoning identified by our action-grounded selection criterion contains information that the policy uses for control.

We further test whether otherwise valid reasoning can be exchanged across different points in a trajectory.

Replacing native reasoning with reasoning from another control state substantially reduces performance by up to 41.8% on LIBERO and by 22.2% on RoboCasa. On LIBERO, this degradation is larger for contrasting-phase than same-phase replacements, while both replacement conditions perform similarly poorly on RoboCasa. Thus, effective control depends not only on generating reasoning, but on generating reasoning whose content is appropriate for the current control state.

D. Real-World Robot Manipulation

Finally, we evaluate AWARE on a physical robot using OpenVLA-ECoT [10] as the underlying VLA backbone. We apply the same offline post-training procedure and deploy the resulting adaptive policy on a WidowX robot. We evaluate four pick-and-place tasks in two environments: banana-to-pink-plate (tabletop), brush-to-blue-towel (tabletop), carrot-to-yellow-plate (toy sink), and eggplant-in-pot (toy sink). We define 3 object layouts per task and execute 8 rollouts per layout, for a total of 24 episodes per task and 96 total episodes per method. Across the resulting 192 real-robot episodes, we show that state-dependent reasoning also emerges during closed-loop real-world control while preserving OpenVLA-ECoT's full-reasoning success. Fig. 5 reports the task success rate across all tasks for both AWARE

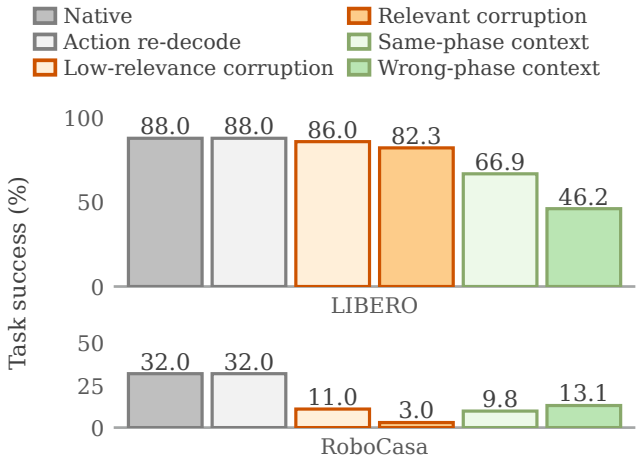


Fig. 7: **Action-relevant reasoning causally affects control.** At a matched reasoning budget, corrupting action-relevant reasoning causes a substantially larger drop in task success than corrupting low-relevance reasoning, while re-decoding unchanged reasoning leaves performance unchanged. Replacing the native trace with reasoning from another control state further degrades performance on both benchmarks.

and the full-reasoning baseline, OpenVLA-ECoT, together with qualitative visualizations of the rollouts illustrating how the retained reasoning primitives change at different stages of manipulation. The average skip rate during deployment reached 40.85%, reducing the total number of generated output tokens by 18.5% on average. This resulted in a 239 ms reduction in reasoning-decoding latency per model call. We further observe that this primitive retention behavior is consistent across both the RoboCasa simulation and the real-world WidowX experiments. As shown in Fig. 6, in RoboCasa, the retention rates of next-motion and object-bounding-box primitives are high during the transport phase, but decreases once the destination is reached. At that point, the gripper path becomes the most highly retained primitive. Despite the different backbone architectures used, MiniVLA for RoboCasa and OpenVLA-ECoT for WidowX, this behavior was observed during real-world deployment (Fig. 5).

We hypothesize that during transport, future-motion and target-grounding reasoning are particularly useful for action prediction, whereas near the destination, motion reasoning becomes less useful and gripper-path reasoning more important for precise placement.

E. Adaptive Reasoning Efficiency

The average efficiency gains only partially capture the behavior of AWARE. As shown in Table II, the amount of reasoning generated by the policy varies substantially across deployment trajectories. We measure the 5th and 95th percentiles of the average skip rates of individual episodes. While the dataset-wide skip rates are moderate, episode-mean skip rates reach 21.8% on LIBERO-90, 54.1% on RoboCasa, and 71.1% on WidowX at the 95th percentile. Thus, AWARE does not simply learn a fixed compression of the full-reasoning trace, but uses very different reasoning budgets across trajectories.

This variation also appears in reasoning-decoding latency on the real-robot inference setup. On the OpenVLA-

Metric (unit)	Full → AWARE	Δ (%)	P5–P95
LIBERO-90 — 900 ep., MiniVLA 0.5B, GH200			
• Primitive skip (%)	11.65	—	2.4–21.8
▼ Reasoning bodies (tok.)	229 → 192.0	▼16.2	140.9–260.2
▼ Total output (tok.)	248 → 213.9	▼13.7	180.8–300.1
▼ Decode reasoning (ms)	537 → 450	▼16.2	330–520
WidowX — 96 ep., OpenVLA-ECoT 7B, H200 TensorRT-LLM			
• Primitive skip (%)	40.85	—	16.3–71.1
▼ Reasoning bodies (tok.)	187 → 135.3	▼27.65	71.8–188.0
▼ Total output (tok.)	281 → 229.1	▼18.5	159.3–321.6
▼ Decode reasoning (ms)	1,343.5 → 1,104.1	▼17.8	811–1,315
RoboCasa — 200 ep., MiniVLA 0.5B, GH200			
• Primitive skip (%)	29.82	—	13.8–54.1
▼ Reasoning bodies (tok.)	108 → 76.3	▼29.4	46.9–98.3
▼ Decode reasoning (ms)	252 → 179	▼29.0	110–230

TABLE II: **AWARE reduces reasoning and decoding cost across models and embodiments.** We report dataset-wide means for Full Reasoning and AWARE, together with the relative reduction. The final column gives the 5th and 95th percentiles of AWARE's per-episode mean, highlighting substantial variation in reasoning cost across trajectories. ▼ denotes reductions relative to Full Reasoning and • the primitive skip rate.

ECoT inference setup , the 5th percentile of episode-mean reasoning-decoding latency is 811 ms, 39.6% below the dataset-wide Full Reasoning mean. This complements the phase-wise analysis above: AWARE adapts not only *which* reasoning primitives it keeps, but also *how much* reasoning it performs over the course of a trajectory for the specific task and scene. The resulting efficiency gains therefore come from changing the reasoning budget during deployment rather than uniformly shortening every trace. Combined with the comparable task success of AWARE and Full Reasoning, these results show that AWARE can substantially reduce reasoning and inference cost while preserving the overall control benefits of embodied chain-of-thought.

V. CONCLUSION

We introduced AWARE, an offline approach that teaches reasoning-capable VLAs to adapt what and how much they reason about at each control step. Using action prediction as a task-grounded signal, AWARE identifies state-dependent reasoning targets and distills them back into the same policy while preserving full-reasoning action behavior. Across simulated and real-world manipulation, AWARE reduces explicit reasoning and inference cost while maintaining the benefits of full reasoning. We further find that reasoning changes systematically throughout execution, and controlled interventions show that action-relevant reasoning causally affects control. Together, our results suggest that effective embodied reasoning is not simply a question of reasoning more or less, but of performing the right reasoning for the current action.

While our experimental results are promising, our approach has several limitations. First, our reasoning selection criterion measures the utility of reasoning through its effect on the current action and may therefore miss reasoning that is only useful over longer horizons. Second, while AWARE can identify which annotated reasoning primitives are useful for control, it does not determine what reasoning

primitives should be defined for a new robot or task. Finally, our objective does not optimize reasoning-decoding latency directly; an interesting direction for future work would be to optimize reasoning decisions against long-horizon task returns, for example through reinforcement learning, while explicitly trading off control performance and inference cost.

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